and now we have the right-hand side of Eq. 10.4. Turning to the left-hand side, we find that

$$\frac{\partial L}{\partial x} = -\omega^2 x. \qquad (10.7)$$

Setting the left and right sides (Eqs. 10.7 and 10.6) of the Lagrange equation equal to each other, we get

$$-\omega^2 x = \ddot{x}. \qquad (10.8)$$

This equation is, of course, equivalent to $F = ma$. Why is there a minus sign? Because the force is a restoring force—its direction is opposite to the direction of the displacement. By now you have seen this type of equation enough to know that the solution contains sines and cosines. The general solution is

$$x = A\cos(\omega t) + B\sin(\omega t), \qquad (10.9)$$

which shows us that ω is indeed the frequency of the oscillator. When we differentiate twice, we pull out a factor of ω^2.

Exercise 10.1: Find the second time derivative of x in Eq. 10.9, and thereby show that it solves Eq. 10.8.

10.2 THE QUANTUM MECHANICAL DESCRIPTION

Now, let's return to our microscopic version of the weight-and-spring system—let's say no bigger than a single molecule. At first, this seems ridiculous. How could we ever build a spring that small? But in fact nature provides all sorts of microscopic springs. Many molecules consist of two atoms—for example, a heavy atom and a light one. There are forces holding the